Homework 10

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1 H 10-1

1.1

The project still costs 80, so the entrepreneur is investing 10 and raising 70 from equityholders. We previously showed that the project has total discounted cashflows of $\frac{0.5\times(140+90)}{1.15} = 100$, so the equityholders should have a stake to $\frac{70}{100} = 70\%$ of cashflows. That puts the equityholder's rate of return at exactly the assumed weighted average cost of capital: $\frac{0.5\times0.7\times(140+90)-70}{70} = 15\%$.

1.2

The equityholder is effectively bearing all the risk since there is no debt being contributed by a bank; therefore, their return should be the unlevered return. It thus makes sense that the equityholders have the same rate of return as the project itself.

2 H 10-2

We are given that the debt-to-equity ratio of 0.4, an equity beta of 1.4, and a debt beta of 0.2. We can interpret the first as $\frac{D}{E} = 0.4$, which means all further ratios will be satisfied if we write $D = 0.4$ and $E = 1$. That lets us solve for the unlevered equity beta:

$$\beta_U = \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D = \frac{1}{1.4}\times 1.4 + \frac{0.4}{1.4}\times 0.2 = 1 + \frac{0.4}{7} = \frac{1.4}{7} = 0.2$$

Now we hold this fixed and reverse-engineer the equity beta of 1.2:

$$0.2 = \frac{1}{1.4} \times 1.2 + \frac{0.4}{1.4}\beta_D \implies \beta_D = \frac{7}{2}\left(0.2 - \frac{5}{7}\right) = \frac{7}{10} - \frac{5}{2} = 0.7 - 2.5$$

so $\boxed{\beta_D = -1.8}$.

3 H 10-3

The weighted average cost of capital is given by

$$r_u = \frac{E}{D + E - C} r_E + \frac{D}{D + E - C} r_D - \frac{C}{D + E - C} r_f$$

Plugging in the numbers, we have

$$r_u = \frac{20}{20 + 40 - 4}(0.04 + 1.2 \times 0.06) + \frac{40}{20 + 40 - 4}(0.04 + 0.1 \times 0.06) - \frac{4}{20 + 40 - 4} \times 0.04$$

which solves to

$$r_u = \frac{20}{56}(0.112) + \frac{40}{56}(0.046) - \frac{4}{56}(0.04) \implies \boxed{r_u = 7.57\%}$$

If we instead use the cash to clear debt and hold the betas constant, we instead can find an unlevered equity beta so that we have a beta for the asset:

$$\boxed{\beta_u = \frac{40}{40 + 16} \times 1.2 + \frac{16}{40 + 16} \times 0.1 = 0.886}$$